2082:

Long Answer Questions: [10 marks each]

1. Distinguish between high frequency emphasis filter and Laplacian filter. Find the output of the following image using histogram equalization, where number of possible gray levels is 8.

3	3	3	3	3
2	3	4	3	2
2	4	4	3	2
2	3	4	3	2
3	3	3	3	3

2. Define noise. Explain the models for image degradation and restoration process.
3. What are the differences between spatial domain and frequency domain? Compute the 2D DFT of the following image.

1	**1**	**1**	**1**
1	**1**	**1**	**1**
1	**1**	**1**	**1**
1	**1**	**1**	**1**

Short Answer Questions: [5 marks each]

4. Discuss about chain code. Write the shape number of the object having chain code {0, 7, 5, 4, 3, 1}.
5. How logic operation on binary images can be used for masking and feature detection? Explain.
6. Define pattern and pattern class. How decision theoretic methods minimize the probability of misclassification?
7. Given the following image, detect the edge using magnitude and direction of gradient, using Prewitt operator.

0	30	60
5	32	62
10	38	64

8. What is image segmentation? How do you detect horizontal and vertical line?

9. Describe the steps in digital image processing.
10. How do you measure distance between pixels? Discuss about region oriented segmentation.
11. Define clipping and contrast stretching. Compute Hadamard transform of the data sequence {1, 2, 0, 3}.
12. Distinguish between forward and inverse transform. Define pixel coding, run length and bit plane.

2081:

Long Answer Questions: [10 marks each]

1. Define linear filter. Given the following histogram of image, stretch to the whole dynamic range using histogram equalization.

Gray Level	0	1	2	3	4	5	6	7
Frequency	0	25	0	10	35	50	0	0

2. Write expression for forward and inverse Discrete Fourier Transform (DFT) for 2D signal. What are the properties of DFT?
3. Briefly list any two noise models. From the following probability distribution construct the Huffman Code for each gray level.

Rk	0	1	2	3	4	5
P(rk)	0.4	0.18	0.1	0.25	0.7	0.05

Short Answer Questions: [5 marks each]

4. Define region and boundary. List some elements of visual perception.
5. What are implications of periodicity and symmetry? Discuss about lossless predictive model.

6. Describe Chain Codes with necessary examples. How can you make chain code rotation and scaling invariant?
7. How vertical lines and horizontal lines are detected? Illustrate with an example.
8. Write the algorithm for basic global thresholding.
9. How do you represent image in spatial domain? Differentiate between intensity level and bit plane slicing.
10. What is pattern recognition? Discuss about Mexican Hat Filters.
11. What are the usages of derivative based filters? Derive mask for Laplacian second order derivative based filter.
12. Show that the points (1,1), (-2,7) and (3,3) are collinear using Hough Transform and find the equation of the line.

2080:

Long Answer Questions: [10 marks each]

1. Explain different steps in digital image processing. Define neighbor of a pixel. How do you find distances between pixels?
2. Why do we need to enhance the image? How do you use histogram in image enhancement? Differentiate between log and power log transformation.
3. Explain the properties of fourier transform. Derive the relation for 1-D fast fourier transformation in spatial domain.

Short Answer Questions: [5 marks each]

4. Explain image degradation and restoration process.
5. Why do we need to remove noise? Discuss about median, max and min filter.
6. Define image segmentation. Explain the three basic types of grey level discontinuities detection.
7. How band reject filters are used to remove periodic noise? Explain.
8. Describe the algorithm for region splitting and merging in brief.
9. Explain how interpixel redundancy can be identified and exploited.
10. Explain erosion with an example.
11. How signature is used in boundary representation? What does shape number mean? Explain.

12. Define pattern and pattern class. How do you detect foreground and background of the image? Explain.